



Zephyr UART Driver: Power Management Implementation

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UART – Default Zephyr Driver

```
DEVICE_DT_INST_DEFINE(n,  
    &ifx_cat1_uart_init##n,  
    NULL,  
    &ifx_cat1_uart##n##_data,  
    &ifx_cat1_uart##n##_cfg,  
    PRE_KERNEL_1,  
    CONFIG_SERIAL_INIT_PRIORITY,  
    &ifx_cat1_uart_driver_api);
```

- Zephyr device instantiation occurs without PM device management
- There is no explicit definition of the callback responsible for managing the device's power consumption.

Infineon UART Deep Sleep Callback

- Infineon's PDL defines a specific callback to handle the SCB UART transitions in and out of Deep Sleep mode.

Cy_SCB_UART_DeepSleepCallback()

```
case CY_SYSPM_CHECK_READY:
```

```
{
```

```
/* Check whether the High-level API is not busy  
executing the transmit or receive operation.*/
```

```
    if {...}
```

```
/* If all data elements were transmitted, the UART is  
ready to enter Deep Sleep mode. */
```

```
    if {...}
```

```
/* Disable the UART. The transmitter stops driving the  
lines and the receiver stops receiving data.*/
```

```
    Cy_SCB_UART_Disable(locBase,  
                        locContext);
```

```
    retStatus = CY_SYSPM_SUCCESS;
```

```
}
```

Infineon UART Deep Sleep Callback

```
Cy_SCB_UART_Disable()
{
    SCB_CTRL(base) &=
(uint32_t)~SCB_CTRL_ENABLED_Msk;
}
```

- Disables the SCB by clearing the **ENABLED** bit in the **SCB_CTRL** register.
- **Software-writable** only.
- **No physical effect** is specified for this bit.

19.1.50 SCB2_CTRL

Description: Generic control

Address: 0x429A0000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	Reserved				RW			
HW Access	Reserved				R			
Name	Reserved [7:4]				OVS [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	RW		Reserved			RW	Reserved	
HW Access	R		Reserved			R	Reserved	
Name	MEM_WIDTH [15:14]		Reserved [13:11]			EZ_MODE	Reserved [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	Reserved							RW
HW Access	Reserved							R
Name	Reserved [23:17]							ADDR_ACCEPT

Bits	31	30	29	28	27	26	25	24
SW Access	RW	Reserved					RW	
HW Access	R	Reserved					R	
Name	ENABLED	Reserved [30:26]					MODE [25:24]	

UART Deep Sleep Callback: Zephyr Implementation

```
static int ifx_cat1_uart_pm_action(*dev, action)
{
    switch (action) {
        case PM_DEVICE_ACTION_SUSPEND:
            if (Cy_SCB_UART_DeepSleepCallback(&cb_params,
                                              CY_SYSPM_CHECK_READY))
                return 0;
    }
}
```



Zephyr's power management callback defined using Infineon's callback.

```
PM_DEVICE_DT_INST_DEFINE(n, ifx_cat1_uart_pm_action);
```



Creation of the structure pointing to the callback.

```
DEVICE_DT_INST_DEFINE(n, &ifx_cat1_uart_init##n,
                      PM_DEVICE_DT_INST_GET(n),
                      &ifx_cat1_uart##n##_data,
                      &ifx_cat1_uart##n##_cfg,
                      PRE_KERNEL_1,
                      CONFIG_SERIAL_INIT_PRIORITY,
                      &ifx_cat1_uart_driver_api);
```



Device instantiation with Zephyr's PM.

Selected Kconfig options

- CONFIG_PM=y
- CONFIG_PM_POLICY_DEFAULT=y
- CONFIG_PM_DEVICE=y
- CONFIG_PM_DEVICE_SYSTEM_MANAGED=y

PM Subsystem Configuration

- CONFIG_INSTRUMENTATION_TRIGGER_FUNCTION="instrumentation_trigger"
- CONFIG_INSTRUMENTATION=y
- CONFIG_INSTRUMENTATION_MODE_CALLGRAPH=y
- CONFIG_INSTRUMENTATION_EXCLUDE_FILE_LIST="....."
- CONFIG_INSTRUMENTATION_EXCLUDE_FUNCTION_LIST="....."
- CONFIG_INSTRUMENTATION_MODE_CALLGRAPH_TRACE_BUFFER_SIZE=32768
- CONFIG_INSTRUMENTATION_MODE_CALLGRAPH_BUFFER_OVERWRITE=n
- CONFIG_INSTRUMENTATION_STOPPER_FUNCTION="instrumentation_stopper"
- CONFIG_INSTRUMENTATION_DYNAMIC_TRIGGER=n

Instrumentation Subsystem Configuration

Main code instrumentation.

```
int main(void)
{
    indicator_active_on();
    k_busy_wait(BLINK_ON_MS * 1000U);
    instrumentation_trigger();
    indicator_active_off();
    k_msleep(SLEEP_BETWEEN_BLINKS_MS);
    instrumentation_stopper();

    while (1) {
        k_busy_wait(100000);
    }
    return 0;
}
```

 One pass to populate the call stack in RAM.

 Infinite loop to read the call stack via UART.

Call Stack: Zephyr Driver Without PM

Thread Name	Thread ID	CPU	Mode	Timestamp	Function(s)

main	0x240bf388	0)	0	204711914 ns	instrumentation_trigger();
main	0x240bf388	0)	0	204833984 ns	indicator_active_off() {
main	0x240bf388	0)	0	204833984 ns	led_set() {
main	0x240bf388	0)	0	204864501 ns	gpio_pin_set_dt() {
main	0x240bf388	0)	0	204895019 ns	gpio_pin_set() {
main	0x240bf388	0)	0	204925537 ns	z_impl_gpio_pin_set() {
main	0x240bf388	0)	0	204956054 ns	z_impl_gpio_pin_set_raw() {
main	0x240bf388	0)	0	204986572 ns	z_impl_gpio_port_clear_bits_raw() {
main	0x240bf388	0)	0	205017089 ns	gpio_ifx_port_clear_bits_raw() {
main	0x240bf388	0)	0	205047607 ns	gpio_ifx_valid_mask();
main	0x240bf388	0)	0	205291748 ns	k_msleep() {
main	0x240bf388	0)	0	205322265 ns	k_sleep()
main	0x240bf388	0)	0	210449218 ns	/* --> Scheduler switched OUT from thread 'main' */
thread-none	none-thread	0)	0	210540771 ns	/* <-- Scheduler switched IN thread 'thread-none' */
thread-none	none-thread	0)	0	210571289 ns	z_thread_entry() {
idle	0x240bf290	0)	7	210723876 ns	idle() {

GPIO call stack.

Entering the idle thread.

Call Stack: Zephyr Driver Without PM

idle	0x240bf290	0)	7		211029052 ns		pm_system_suspend() {
idle	0x240bf290	0)	7		211303710 ns		pm_policy_next_event_ticks();
idle	0x240bf290	0)	7		211425781 ns		pm_policy_next_state()
idle	0x240bf290	0)	7		211853027 ns		pm_suspend_devices() {
idle	0x240bf290	0)	7		211883544 ns		z_device_get_all_static();
idle	0x240bf290	0)	7		212341308 ns		pm_state_set() {
idle	0x240bf290	0)	7		212371826 ns		enter_cpu_sleep() {
idle	0x240bf290	0)	7		212402343 ns		indicator_cpu_sleep_on() {
idle	0x240bf290	0)	7		212890625 ns		pm_irq_prologue();
idle	0x240bf290	0)	7		212951660 ns		Cy_SysPm_CpuEnterSleep() {
idle	0x240bf290	0)	7		214813232 ns		indicator_cpu_sleep_off() {
idle	0x240bf290	0)	7		215362548 ns		pm_system_resume() {
idle	0x240bf290	0)	7		215576171 ns		pm_resume_devices();
idle	0x240bf290	0)	7		215637207 ns		pm_state_exit_post_ops()
idle	0x240bf290	0)	7		219299316 ns		pm_state_notify()
idle	0x240bf290	0)	7		221923828 ns		/* --> Scheduler switched OUT from thread 'idle' */
main	0x240bf388	0)	0		222015380 ns		/* <-- Scheduler switched IN thread 'main' */
main	0x240bf388	0)	0		222442626 ns		instrumentation_stopper();

Low-power state evaluation
and direct transition into it.

Exiting low-power
state and switching
to the main thread.

Call Stack: Zephyr Driver with PM

Thread Name	Thread ID	CPU	Mode	Timestamp	Function(s)

main	0x240bf3f0	0)	0	204742431 ns	instrumentation_trigger();
main	0x240bf3f0	0)	0	205352783 ns	k_msleep() {
main	0x240bf3f0	0)	0	205383300 ns	k_sleep() {
main	0x240bf3f0	0)	0	205413818 ns	z_impl_k_sleep()
main	0x240bf3f0	0)	0	210540771 ns	/* --> Scheduler switched OUT from thread 'main' */
thread-none	none-thread	0)	0	210601806 ns	configure_builtin_stack_guard();
thread-none	none-thread	0)	0	210662841 ns	/* <-- Scheduler switched IN thread 'thread-none' */
thread-none	none-thread	0)	0	210693359 ns	z_thread_entry() {
idle	0x240bf2f8	0)	7	210845947 ns	idle() {
idle	0x240bf2f8	0)	7	211151123 ns	pm_system_suspend() {
idle	0x240bf2f8	0)	7	211395263 ns	pm_policy_next_event_ticks();
idle	0x240bf2f8	0)	7	211517333 ns	pm_policy_next_state()

Low-power state
evaluation

Entering the idle
thread.

Call Stack: Zephyr Driver with PM

idle	0x240bf2f8	0)	7		211944580 ns		pm_suspend_devices() {
idle	0x240bf2f8	0)	7		212371826 ns		ifx_cat1_uart_pm_action() {
idle	0x240bf2f8	0)	7		212402343 ns		Cy_SCB_UART_DeepSleepCallback() {
idle	0x240bf2f8	0)	7		212432861 ns		Cy_SCB_UART_GetTransmitStatus();
idle	0x240bf2f8	0)	7		212493896 ns		Cy_SCB_UART_GetReceiveStatus();
idle	0x240bf2f8	0)	7		212554931 ns		Cy_SCB_UART_IsTxComplete();
idle	0x240bf2f8	0)	7		212799072 ns		Cy_SCB_UART_GetNumInRxFifo();
idle	0x240bf2f8	0)	7		212921142 ns		Cy_SCB_UART_Disable();
idle	0x240bf2f8	0)	7		214233398 ns		pm_state_set() {
idle	0x240bf2f8	0)	7		214263916 ns		enter_cpu_sleep() {
idle	0x240bf2f8	0)	7		214294433 ns		indicator_cpu_sleep_on() {
idle	0x240bf2f8	0)	7		214813232 ns		pm_irq_prologue();
idle	0x240bf2f8	0)	7		214874267 ns		Cy_SysPm_CpuEnterSleep() {
idle	0x240bf2f8	0)	7		216766357 ns		indicator_cpu_sleep_off() {
idle	0x240bf2f8	0)	7		217346191 ns		pm_system_resume() {
idle	0x240bf2f8	0)	7		217559814 ns		pm_resume_devices() {
idle	0x240bf2f8	0)	7		217620849 ns		ifx_cat1_uart_pm_action() {
idle	0x240bf2f8	0)	7		217651367 ns		Cy_SCB_UART_DeepSleepCallback() {
idle	0x240bf2f8	0)	7		217681884 ns		Cy_SCB_UART_Enable();
idle	0x240bf2f8	0)	7		221954345 ns		k_sched_unlock() {
idle	0x240bf2f8	0)	7		224578857 ns		/* --> Scheduler switched OUT from thread 'idle' */
main	0x240bf3f0	0)	0		224609375 ns		configure_builtin_stack_guard();
main	0x240bf3f0	0)	0		224700927 ns		/* <-- Scheduler switched IN thread 'main' */
main	0x240bf3f0	0)	0		225128173 ns		instrumentation_stopper();

Peripheral suspension via
registered PM and transition
into low-power state.

Exiting low-power mode,
resuming suspended devices,
and switching to the main thread.

Modified Infineon UART Deep Sleep Callback

```
Cy_SCB_UART_DeepSleepCallback{
```

```
case CY_SYSPM_CHECK_READY:
```

```
{
```

```
    scb_enabled_before =  
(SCB_CTRL(locBase) >> 31U) & 1U;
```



Reading the **ENABLED** bit before disabling.

```
        Cy_SCB_UART_Disable(locBase,  
locContext);
```

```
    scb_enabled_after =  
(SCB_CTRL(locBase) >> 31U) & 1U;  
}
```



Reading the **ENABLED** bit after disabling.

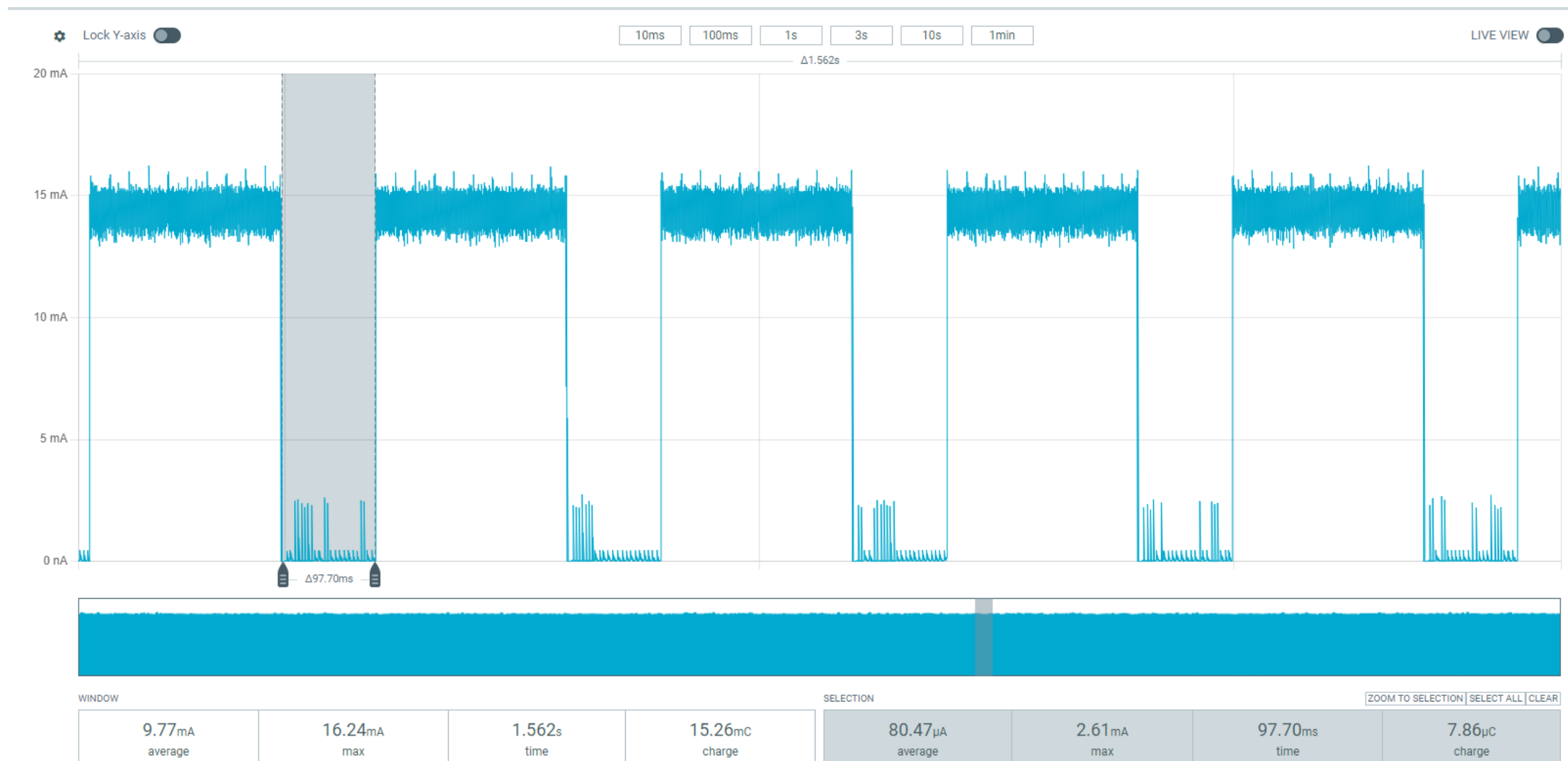
```
}
```

SCB_CTRL Register Read:

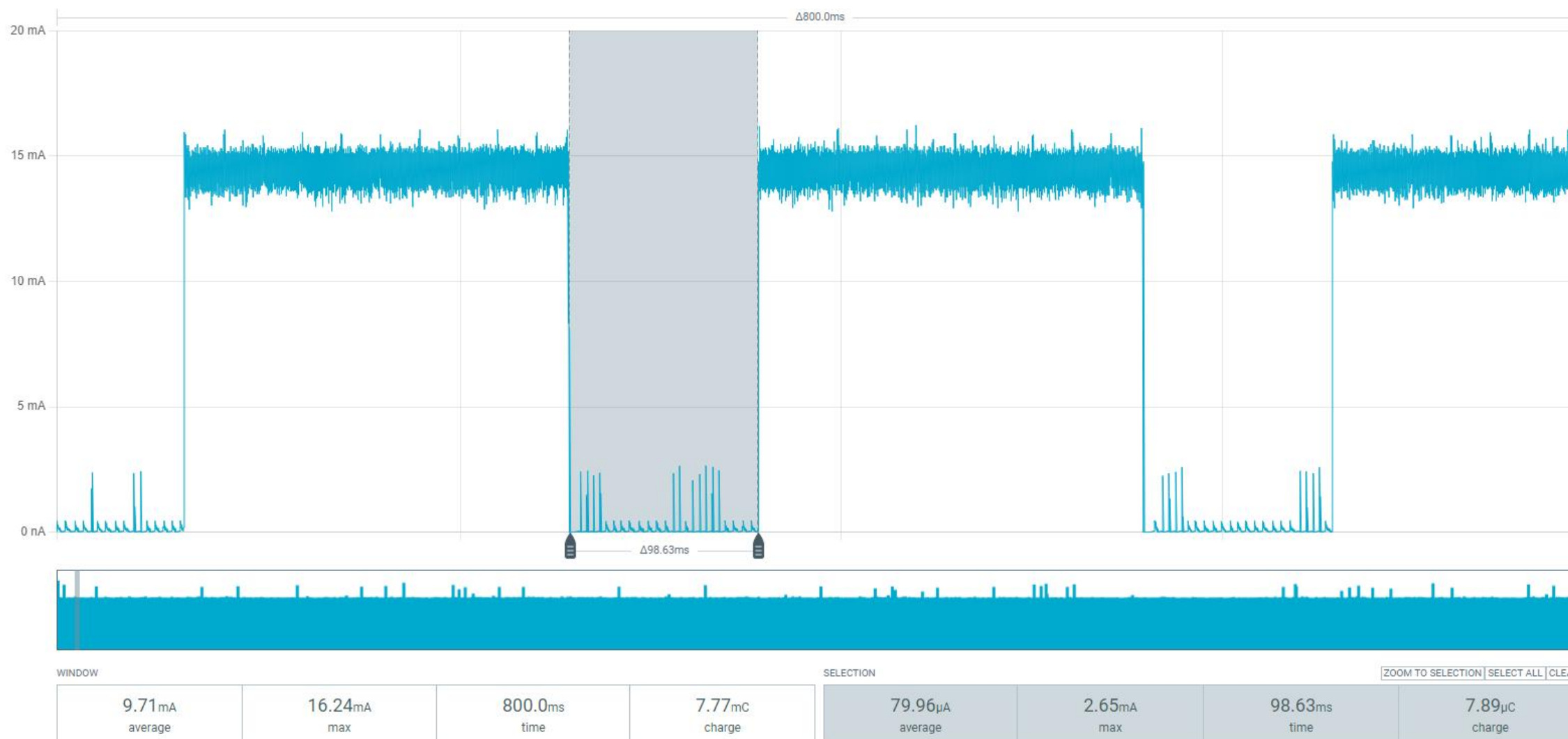
	No Device PM	PM Device Enabled
SCB_CTRL.Enabled Before Transition	Null	1
SCB_CTRL.Enabled After Transition	Null	0

- The ENABLED bit is modified when running the power management callback defined in the Zephyr driver.

Current Consumption Measurement: Device without PM

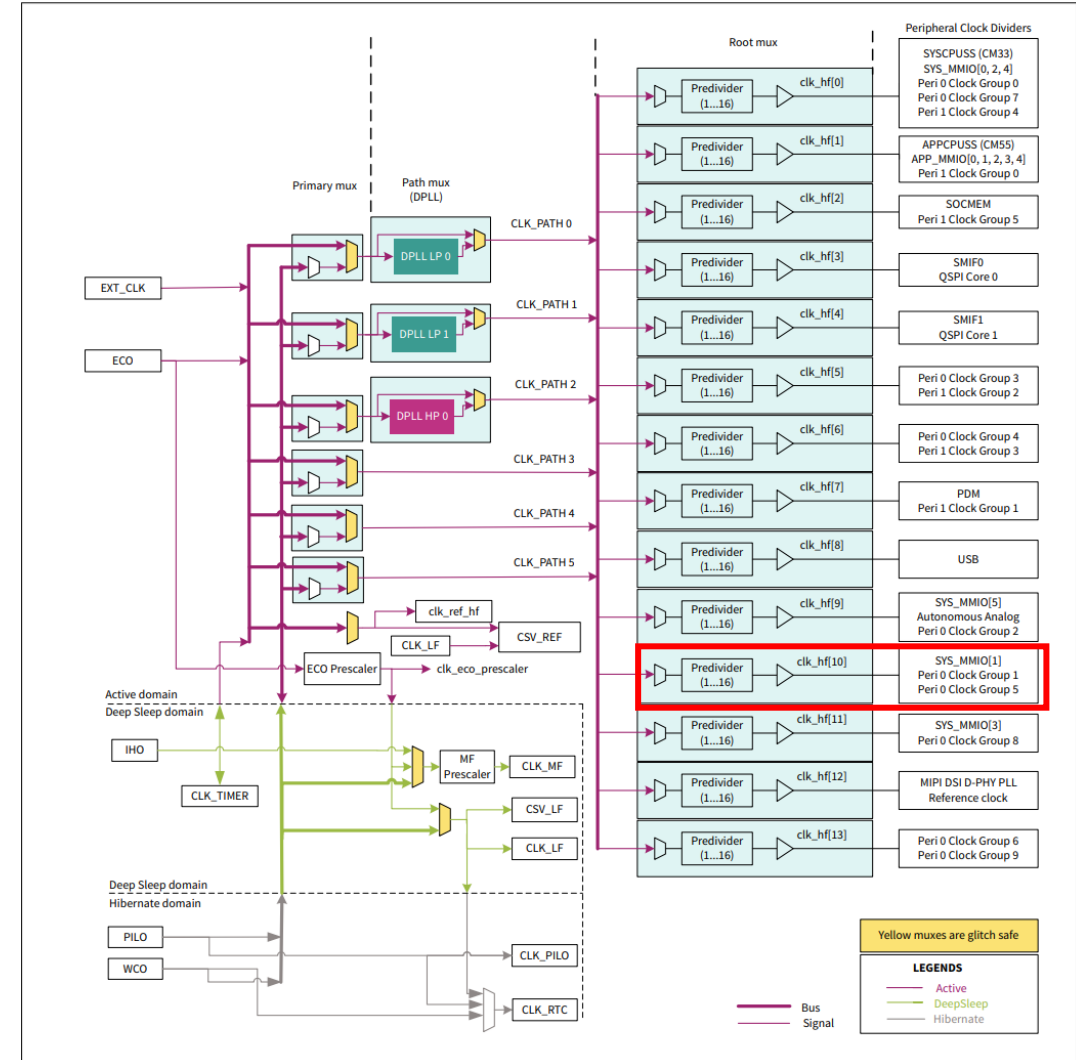


Current Consumption Measurement: Device with PM



Clock Gating Strategy

- Disable High-Frequency Clock [10].
- Clocked peripherals:
 - GPIO
 - SCB
 - TCPWM



Clock Gating Strategy

- Set the **ENABLE** bit in the **CLK_ROOT_SELECT** register
- Write via Secure Mode

Description: Clock Root Select Register

Address: 0x42401240

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	Reserved				RW			
HW Access	Reserved				R			
Name	Reserved [7:4]				ROOT_MUX [3:0]			
Bits	15	14	13	12	11	10	9	8
SW Access	Reserved				RW			
HW Access	Reserved				R			
Name	Reserved [15:12]				ROOT_DIV [11:8]			
Bits	23	22	21	20	19	18	17	16
SW Access	Reserved							
HW Access	Reserved							
Name	Reserved [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	Reserved						
HW Access	A	Reserved						
Name	ENABLE	Reserved [30:24]						

Bits	Name	Description
31	ENABLE	Enable for this clock root. All clock roots default to disabled (ENABLE==0) except HFCLK0, which cannot be disabled. Default Value: 0

Gating-Free CPU Sleep



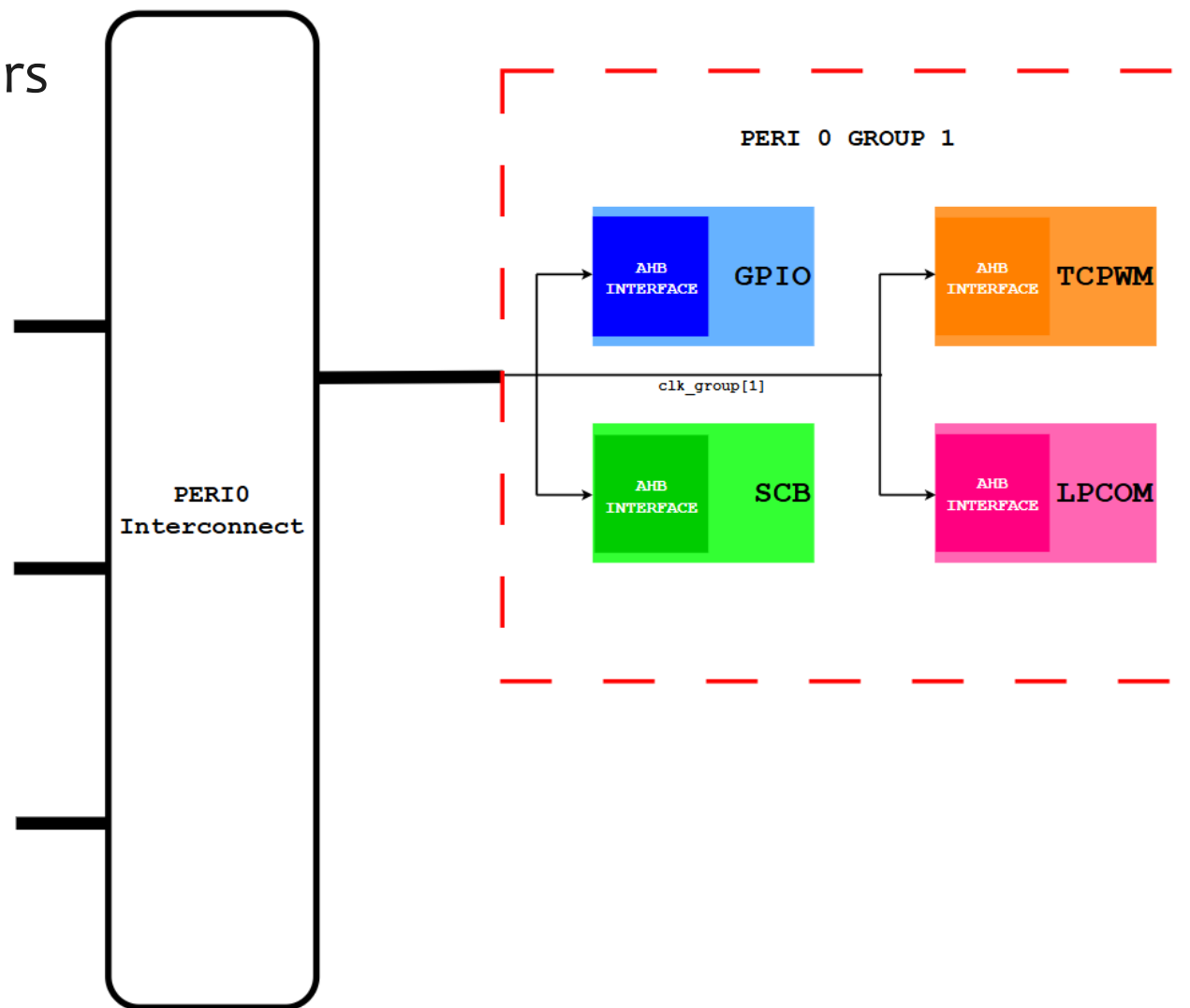
Clock-Gated CPU Sleep



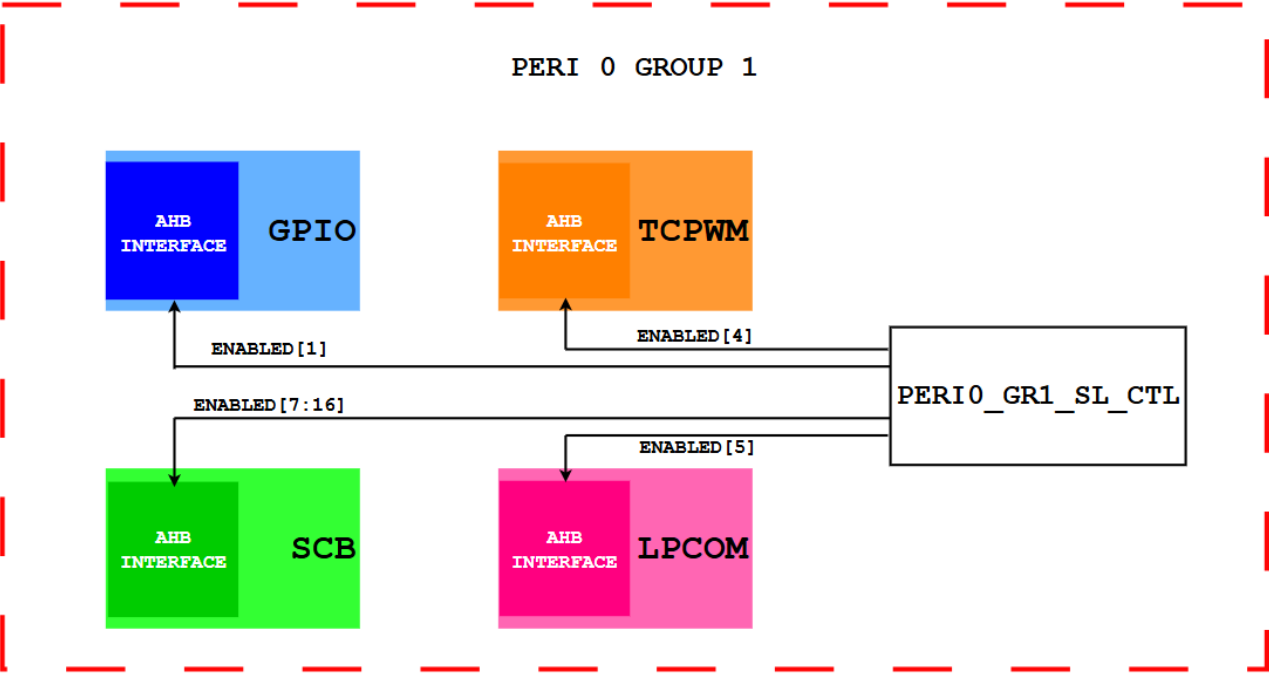
Clock Gating Strategy

Peri interconnect connects AHB masters to groups of peripheral devices.

Slave	Slave Number
GPIO	1
TCPWM	4
LPCOMP	5
SCB[2:11]	7 – 16



Clock Gating Strategy



9.1.7 PERIO_GR1_SL_CTL

Description: Slave control

Address: 0x42004050

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	ENABLED [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	ENABLED [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	Reserved						RW	
HW Access	Reserved						R	
Name	Reserved [23:18]						ENABLED [17:16]	
Bits	31	30	29	28	27	26	25	24
SW Access	Reserved							
HW Access	Reserved							
Name	Reserved [31:24]							

Slave Enable. Each bit indicates whether the respective slave is enabled. If the slave is disabled, its clock is gated off (constant "0").

